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Studierichting: Informatica
Oefeningenreeks 4A nr. 14.8

$$\int x^2 \ln^2 x dx$$

$$\begin{cases} u = \ln^2 x \\ dv = x^2 dx \end{cases} \Rightarrow \begin{cases} du = 2\ln x \cdot \dfrac{1}{x} dx \\ v = \dfrac{1}{3} x^3 \end{cases}$$

$$= \ln^2 x \cdot \frac{1}{3}x^3 - \int \frac{1}{3}x^3 \cdot 2\ln x \cdot \frac{1}{x} dx$$

$$= \ln^2 x \cdot \frac{1}{3}x^3 - \frac{2}{3} \int \ln x \cdot \frac{x^3}{x} dx$$

$$= \ln^2 x \cdot \frac{1}{3}x^3 - \frac{2}{3} \int x^2 \ln x dx$$

$$\begin{cases} u = \ln x \\ dv = x^2 dx \end{cases} \Rightarrow \begin{cases} du = \dfrac{1}{x} dx \\ v = \dfrac{1}{3} x^3 \end{cases}$$

$$= \ln^2 x \cdot \frac{1}{3}x^3 - \frac{2}{3} \cdot \left[\ln x \cdot \frac{1}{3}x^3 - \int \frac{1}{3}x^3 \cdot \frac{1}{x} dx \right]$$

$$= \ln^2 x \cdot \frac{1}{3}x^3 - \frac{2}{3} \cdot \left[\ln x \cdot \frac{1}{3}x^3 - \frac{1}{3} \int x^2 dx \right]$$

$$= \ln^2 x \cdot \frac{1}{3}x^3 - \frac{2}{3} \cdot \left[\ln x \cdot \frac{1}{3}x^3 - \frac{1}{3} \cdot \frac{1}{3}x^3 \right] + C$$

$$= \ln^2 x \cdot \frac{1}{3}x^3 - \frac{2}{3} \cdot \left[\ln x \cdot \frac{1}{3}x^3 - \frac{1}{9}x^3 \right] + C$$

$$= \ln^2 x \cdot \frac{1}{3}x^3 - \frac{2}{9}x^3 \ln x + \frac{2}{27}x^3 + C$$

References